

A Wideband Rat-race Coupler Using Stepped Impedance Resonator

Anuj Kumar Sahoo, *Student Member, IEEE*, Karun Rawat, *Senior member, IEEE*
Indian Institute of Technology, Roorkee, India

Abstract— A wideband rat-race coupler using stepped impedance resonator (SIR) has been proposed in this paper. The proposed coupler operates over frequency range of 1.70 GHz to 2.40 GHz with 35% fractional bandwidth, centered at 2.0 GHz. The measured amplitude and phase imbalances between the two out-of-phase output ports are within 0.5 dB and $180 \pm 10^\circ$. To achieve this, each branch of a conventional rat-race coupler is replaced by the SIRs. The values of the impedances of the lines are computed mathematically first and later realized using microstrip lines. The proposed design offers 75% more wider bandwidth than the conventional rat race coupler, without increasing the overall foot-print. The measured reflection co-efficient is below -10 dB with isolation better than 20 dB in the desired band. The prototype has been fabricated and measured and the results are compared with the simulated data to validate the theory.

Index Terms— coupler, microwave circuits, rat-race, stepped impedance resonator, wideband.

I. INTRODUCTION

Rat-race couplers find application in balanced mixers, differentially fed antenna etc. [1]-[3]. The rat-race is inherently narrowband in nature, owing to the $\lambda/4$ and $3\lambda/4$ transmission lines used in its branches which are designed for a particular frequency. Therefore, several efforts have been carried out in the past to enhance the bandwidth of rat-race coupler [4]-[9]. The basic principle to augment the bandwidth of a coupler is to use multiple sections of it [1]. Nevertheless, a few alternatives have been reported in the past. In [4], the author has replaced the branch realized by $3\lambda/4$ transmission lines of the rat-race, by two short-circuited $\lambda/4$ lines with gap coupling, resulting into band-width more than 20%. Kuo et al. in [5] have proposed a miniaturized rat-race coupler with harmonic suppression using periodic stepped impedance ring resonators. A rat-race coupler explained in [6] has 50% bandwidth, but it comprises of two sections resulting into larger size and therefore requires miniaturization scheme. The work in [5], [7] have mainly focused on miniaturization, rather than wideband characteristics. A compact coplanar waveguide (CPW) phase inverter integrated rat-race coupler is reported [8] with 65% Fractional Bandwidth. In this case, the rat-race coupler is realized utilizing Glass Integrated Passive Device (GIPD) technology. The design in [4] is modified by adding two more short-circuited stubs in [9], which further improved the bandwidth up to 70%.

In this paper, the $\lambda/4$ and $3\lambda/4$ transmission lines of the regular rat-race has been replaced by non-uniform line resonator or stepped impedance resonator (SIR). The methodology proposed in this paper offers 75% more wider bandwidth as compared to conventional rat-race coupler without increasing the

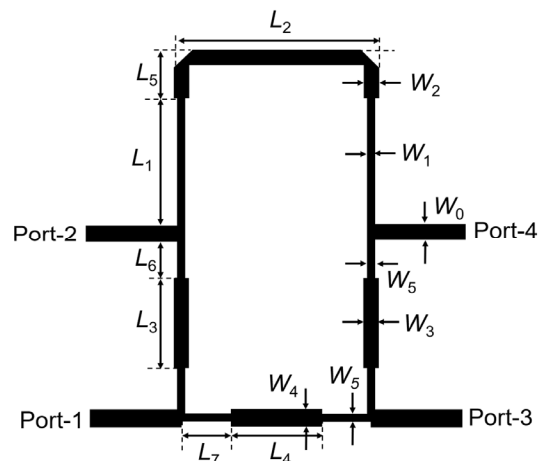


Fig. 1. Topology of the proposed Rat-race coupler.

size. The paper has been organized as follows: section II outlines the necessary theory and design curves. The results and subsequent discussions are included in section III followed by conclusion in section IV.

II. THEORY AND DESIGN TECHNIQUE

Fig. 1 shows the topology of the proposed rat-race coupler. For input given at port 1, the design parameters in terms of impedance and electrical length of each SIR is obtained by keeping $|S_{21}-S_{31}| \leq 0.5\text{dB}$ and $\angle S_{21}-\angle S_{31} \leq 10^\circ$ respectively throughout the band of operation. Similarly, respective goals $|S_{24}-S_{34}| \leq 0.5\text{dB}$ and $\angle S_{24}-\angle S_{34} \leq 180^\circ \pm 10^\circ$ are set for the case when input is given at port 4 and two out-of-phase outputs are measured at port 2 and 3 as shown in Fig. 1. By virtue of the SIRs, the insertion loss and spurious frequency response can be controlled with variation of the impedance ratio of the high and low impedance sections. Hence, it is evident that by employing this technique the insertion loss between the ports of a rat-race can be controlled to yield a wider bandwidth.

A. Background Theory

In order to obtain desired insertion losses and phase difference between different ports, the topology in Fig. 1 is split around the plane of symmetry and is replaced by open and short terminations for even and odd mode analysis, respectively. The subsequent analysis is as follows:

$$Z_{le,1} = -jZ_4 \cot \theta \quad (1a)$$

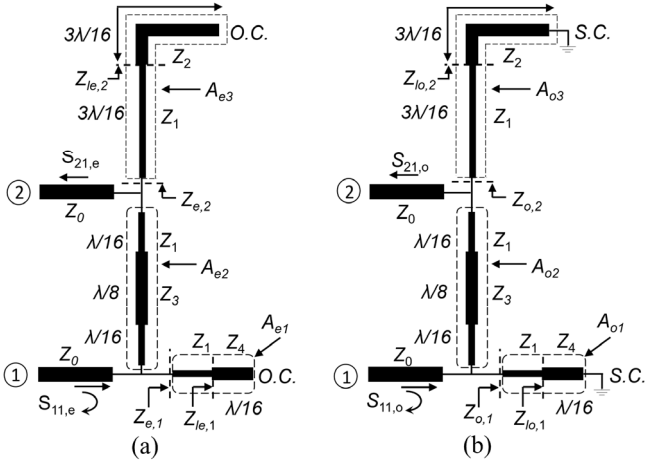


Fig. 2. Even-odd mode analysis of the rat-race coupler (a) Even mode (b) Odd mode.

$$Z_{e1} = Z_1 \left(\frac{Z_{le,1} + jZ_1 \tan \theta}{Z_1 + jZ_{le,1} \tan \theta} \right) \quad (1b)$$

$$Z_{le,2} = -jZ_4 \cot 3\theta \quad (2a)$$

$$Z_{e2} = Z_1 \left(\frac{Z_{le,2} + jZ_1 \tan 3\theta}{Z_1 + jZ_{le,2} \tan 3\theta} \right). \quad (2b)$$

Where Z_{e1} & Z_{e2} are the input impedances seen towards the open-end termination as described in Fig. 2(a). The electric length $\theta = \beta l$, where $l = \lambda/16$. The $Z_{le,1}$ and $Z_{le,2}$ are the intermediate impedances of $\lambda/16$ and $3\lambda/16$ open transmission line sections respectively as shown in Fig. 2(a). From the $ABCD$ matrix of individual branches (A_{e1} , A_{e2} & A_{e3}), the overall $ABCD$ matrix A_e between port 1 and 2 for even mode can be obtained as:

$$A_e = A_{e1} \times A_{e2} \times A_{e3} \quad (3)$$

The overall $ABCD$ matrix of (3) can be used to obtain $S_{11,e}$, $S_{21,e}$ [1]. Similar analysis can be carried out for odd-mode using Fig. 2(b). This will lead to the derivation of overall $ABCD$ matrix A_o in case of odd mode from which one can obtain $S_{11,o}$ & $S_{21,o}$ [1]. Finally, the overall S-parameters of the rat-race coupler can be computed using [1]:

$$S_{11} = 0.5 \times (S_{11,e} + S_{11,o}) \quad (4a)$$

$$S_{21} = 0.5 \times (S_{21,e} + S_{21,o}) \quad (4b)$$

$$S_{31} = 0.5 \times (S_{11,e} - S_{11,o}) \quad (4c)$$

$$S_{41} = 0.5 \times (S_{21,e} - S_{21,o}) \quad (4d)$$

The above analysis considers the input at the sum port, i.e. port 1. A similar analysis can be performed for the rat-race driven at the difference port, i.e. port 4. This will complete the S-matrix of the rat-race coupler.

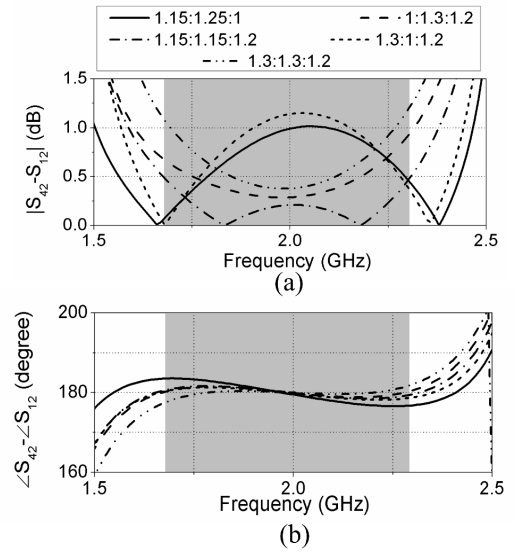


Fig. 3. Comparison between computed result of (a) amplitude imbalance (b) Phase imbalance for various combinations of $k_2:k_3:k_4$.

B. Design Methodology

The intention of this design approach is to obtain the impedances of the SIRs replacing the branches of rate-race coupler in such a way that, the amplitude and phase imbalance and should not exceed 0.5 dB and $180^\circ \pm 10^\circ$ over the desired bandwidth. The impedance Z_1 of SIR as shown in Fig. 2 is set as 70.7Ω , which is same as conventional rate-race coupler design. However, the impedances Z_2 , Z_3 , Z_4 of SIRs in Fig. 2 are calculated with the goal of keeping the amplitude and phase imbalance within the desired limits. The impedance ratio k_2 , k_3 & k_4 , are the respective ratios of impedances Z_2 , Z_3 and Z_4 w.r.t the reference impedance Z_0 . Fig. 3 shows the amplitude and phase imbalance between the transmission parameters S_{42} and S_{12} , for different values of k_2 , k_3 & k_4 , when port 2 is excited. Looking at the results in Fig. 3(a) & (b), it is observed that a trade-off must be made between the bandwidth and maximum allowable amplitude and phase imbalances. The results, are found to be optimum for $k_2:k_3:k_4=1.15:1.25:1.0$. Once, the optimum values were obtained after numerical computation, the SIRs were implemented using microstrip lines. Fig. 4 shows the fabricated prototype which is printed

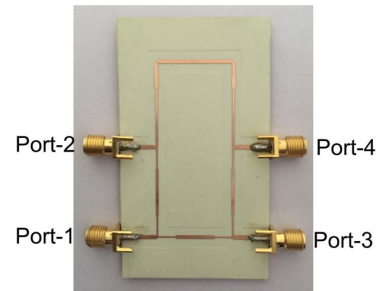


Fig. 4. Photograph of the fabricated prototype.

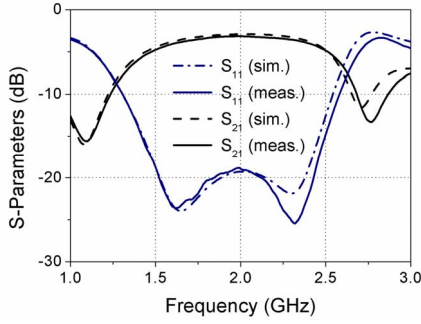


Fig. 5. Simulate and measured S_{11} & S_{21} plot of proposed rat-race coupler.

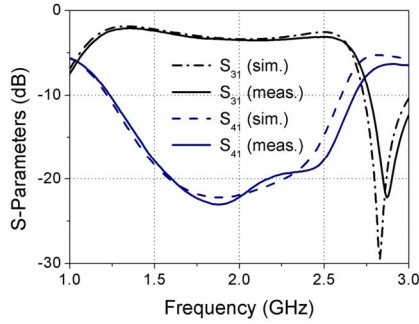


Fig. 6. Simulate and measured S_{31} & S_{41} plot of proposed rat-race coupler

on Rogers RO4350B substrate. The substrate has dielectric constant of 3.66 and height of 0.508 mm, respectively. A slight tuning was done to get best results in electromagnetic simulations. The normalized impedances k_2 , k_3 , k_4 finally used after electromagnetic simulation are 1.06, 1.21 & 1.0, respectively which is in close agreement with the calculated values. The optimized dimensions (in mm) of the design as shown in Fig.1 are as follows: $L_1=16.05$, $L_2=22.6$, $L_3=13.6$, $L_4=11.0$, $L_5=7.62$, $L_6=5.6$, $L_7=5.33$, $W_0=1.09$, $W_1=0.58$, $W_2=0.92$, $W_3=0.71$, $W_4=1.01$, $W_5=0.4$.

III. RESULTS AND DISCUSSION

Measurements are done using Agilent's N5242A Network Analyzer. The measured and simulated S-parameters for the coupler driven by port 1 is presented in Fig. 5 & 6. In both the cases, the measured results seems to follow the simulated values. Isolation between port 1 & 4 is below -18 dB in the band of 1.7 to 2.4 GHz, Fig. 6. The insertion loss between port 1 & 2 and port 1 & 3 varies between 2.9-3.4 dB within this band, Fig. 5. Simulated and measured amplitude imbalance (in dB) and phase imbalance (in degree) are illustrated in Fig. 7. One can see from Fig. 7 (a) that the amplitude imbalance is below 0.5 dB for band of 400 MHz ranging from 1.8-2.2 GHz in case of conventional rat-race coupler. However, for same amplitude imbalance, proposed design operates over frequency 1.7-2.4 GHz resulting into the bandwidth of 700 MHz. Thus, the bandwidth has increased by 75%, compared the conventional approach. The phase imbalance in this region is also below

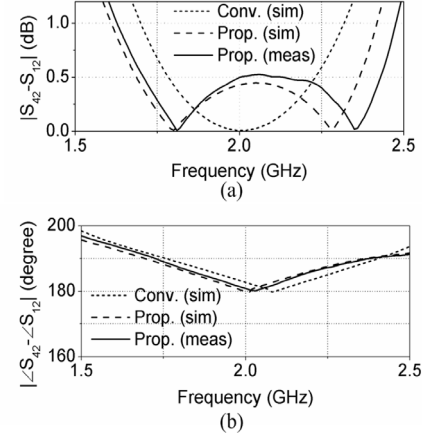


Fig. 7. Comparison between IL difference and Phase difference between conventional and proposed rat-race.

190°. By allowing amplitude imbalance up to 1dB and phase imbalance up to 15°, the fractional bandwidth of the proposed coupler is around 40%.

IV. CONCLUSION

The design scheme is proposed to enhance bandwidth of the rat-race coupler using SIRs while keeping the amplitude & phase imbalance within tight limits. The circuit is simple, planar, and cost effective and doesn't require any additional circuitry or footprint.

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